CURRENT

BS in Chemical Engineering

Degree Requirements

To earn a Bachelor of Science in Chemical Engineering degree from UIC, students need to complete University, college, and department degree requirements. The Department of Chemical Engineering degree requirements are outlined below. Students should consult the *College of Engineering* section for additional degree requirements and college academic policies.

BS in Chemical Engineering Degree Requirements	**Hours**
Nonengineering and General Education Requirements	74
Required in the College of Engineering	48
Technical Elective	3*
Electives outside the Major Rubric	3*
Total Hours—BS in Chemical Engineering	**128**

** Students in the Biochemical Engineering Concentration take a minimum of 8 hours of electives and 130 hours for the degree; see below.*

Nonengineering and General Education Requirements

Courses	**Hours**
ENGL 160— Academic Writing I: Writing for Academic and Public Contexts	3
ENGL 161— Academic Writing II: Writing for Inquiry and Research	3
Exploring World Cultures course[a]	3
Understanding the Creative Arts course[a]	3
Understanding the Past course[a]	3
Understanding the Individual and Society course[a]	3
Understanding U.S. Society course[a]	3
MATH 180—Calculus I[b]	5
MATH 181—Calculus II[b]	5
MATH 210—Calculus III[b]	3
MATH 220—Introduction to Differential Equations I	3
PHYS 141—General Physics I (Mechanics)[b]	4
PHYS 142—General Physics II (Electricity and Magnetism)[b]	4
CHEM 112—General College Chemistry I[b]	5
CHEM 114—General College Chemistry II[b]	5
CHEM 222—Analytical Chemistry	4
CHEM 232—Organic Chemistry I	4
CHEM 233—Organic Chemistry Laboratory I	1
CHEM 234—Organic Chemistry II	4
CHEM 342—Physical Chemistry I	3
CHEM 346—Physical Chemistry II	3
Total Hours—Nonengineering and General Education Requirements	**74**

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BS in Chemical Engineering Degree Requirements	**Hours**
Nonengineering and General Education Requirements	73
Required in the College of Engineering	48
Technical Elective	3*
Electives outside the Major Rubric	4*
Total Hours—BS in Chemical Engineering	**128**

** Students in the Biochemical Engineering Concentration take a minimum of 8 hours of electives and 130 hours for the degree; see below.*

Nonengineering and General Education Requirements

Courses	**Hours**
ENGL 160— Academic Writing I: Writing for Academic and Public Contexts	3
ENGL 161— Academic Writing II: Writing for Inquiry and Research	3
Exploring World Cultures course[a]	3
Understanding the Creative Arts course[a]	3
Understanding the Past course[a]	3
Understanding the Individual and Society course[a]	3
Understanding U.S. Society course[a]	3
MATH 180—Calculus I[b]	4
MATH 181—Calculus II[b]	4
MATH 210—Calculus III[b]	3
MATH 220—Introduction to Differential Equations I	3
PHYS 141—General Physics I (Mechanics)[b]	4
PHYS 142—General Physics II (Electricity and Magnetism)[b]	4
CHEM 122—General Chemistry I Lecture[c]	4
CHEM 123—General Chemistry Laboratory I[bc]	1
CHEM 124—General Chemistry II Lecture[c]	4
CHEM 125—General Chemistry Laboratory II[bc]	1
CHEM 222—Analytical Chemistry	4
CHEM 232—Organic Chemistry I	4
CHEM 233—Organic Chemistry Laboratory I	2
CHEM 234—Organic Chemistry II	4

[a] *Students should consult the General Education section of the catalog for a list of approved courses in this category.*
[b] *This course is approved for the Analyzing the Natural World General Education category.*

Required in the College of Engineering

Courses	**Hours**
ENGR 100—Orientation[a]	0a
CHE 201—Introduction to Thermodynamics	3
CHE 205—Computational Methods in Chemical Engineering	3
CHE 210—Material and Energy Balances	4
CHE 301—Chemical Engineering Thermodynamics	3
CHE 311—Transport Phenomena I	3
CHE 312—Transport Phenomena II	3
CHE 313—Transport Phenomena III	3
CHE 321—Chemical Reaction Engineering	3
CHE 341—Chemical Process Control	3
CHE 381—Chemical Engineering Laboratory I	2
CHE 382—Chemical Engineering Laboratory II	2
CHE 396—Senior Design I	4
CHE 397—Senior Design II	3
CME 260—Properties of Materials	3
ECE 210—Electrical Circuit Analysis	3
CHE 499—Professional Development Seminar	0
CS 109—C/C++ Programming for Engineers with MatLab	3
Total Hours—Required in the College of Engineering	**48**

[a] *ENGR 100 is one-semester-hour course, but the hour does not count toward the total hours required for graduation.*

Technical Elective

Courses	**Hours**
One technical elective to be chosen from the following list of design-oriented courses: [a]	3
CHE 392—Undergraduate Research (3)[b]	
CHE 410—Transport Phenomena (3)	
CHE 413—Introduction to Flow in Porous Media (3)	
CHE 421—Combustion Engineering (3)	
CHE 422—Biochemical Engineering (3)	
CHE 423—Catalytic Reaction Engineering (3)	
CHE 431—Numerical Methods in Chemical Engineering (3)	
CHE 433—Process Simulation with Aspen Plus (3)	
CHE 438—Computational Molecular Modeling (3)	

CHEM 342—Physical Chemistry I	3
CHEM 346—Physical Chemistry II	3
Total Hours—Nonengineering and General Education Requirements	**73**

[a] *Students should consult the General Education section of the catalog for a list of approved courses in this category.*
[b] *This course is approved for the Analyzing the Natural World General Education category.*
[c] *General Education credit is given for successful completion of both CHEM 122/123 or CHEM 124/125.*

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CHE 440—Non-Newtonian Fluids (3) CHE 441—Computer Applications in Chemical Engineering (3) CHE 445—Mathematical Methods in Chemical Engineering (3) CHE 450—Air Pollution Engineering (4) CHE 456—Fundamentals and Design of Microelectronics Processes (3) CHE 494—Selected Topics in Chemical Engineering (3) **Total Hours—Technical Elective** **3** [a] *Possible technical elective credit for a 400-level CHE course not listed above will require departmental approval by petition to the Undergraduate Committee.* [b] *An appropriate design-related research project may be selected with the approval of the Department of Chemical Engineering.*	**SAME**
Electives outside the Major Rubric **Courses** — **Hours** Electives outside the CHE rubric — 3 **Total Hours—Electives outside the Major Rubric** — **3**	Electives outside the Major Rubric **Courses** — **Hours** Electives outside the CHE rubric — 4 **Total Hours—Electives outside the Major Rubric** — **4**
BS in Chemical Engineering—Biochemical Engineering Concentration	
Students in this concentration complete the following: **Required Courses—Biochemical Engineering Option** — **Hours** ***One technical elective from Chemical Engineering:*** CHE 422—Biochemical Engineering — 3 ***Two electives in nonmajor rubric category from among the following:*** — 5–7 BIOS 350—General Microbiology (3) BIOS 351—Microbiology Laboratory (2) CHEM 352—Introductory Biochemistry (3) CHEM 452—Biochemistry I (4) **Total Hours—Required Courses Biochemical Engineering Concentration[a]** — **8–10** [a] *Due to structure of the concentration and the prerequisites required for some of the courses, students in the concentration will be required to take a minimum of 130 semester hours for the degree.*	**SAME**

Minor in Chemical Engineering	
For the minor, 16–18 semester hours are required, excluding prerequisite courses. Students outside the Department of Chemical Engineering who wish to minor in Chemical Engineering must complete the following:	**SAME**
Prerequisite Courses—Chemical Engineering Minor — **Hours** Choose one of the following courses: 5 CHEM 112—General College Chemistry I (5) CHEM 116—Honors General Chemistry I (5) CHEM 342—Physical Chemistry I 3 CS 109—C/C++ Programming for Engineers with MatLab 3 MATH 180—Calculus I 5 MATH 181—Calculus II 5 MATH 210—Calculus III 3 MATH 220—Introduction to Differential Equations 3 PHYS 141—General Physics I (Mechanics) 4 PHYS 142—General Physics II (Electricity and Magnetism) 4 **Total Hours—Prerequisites for Chemical Engineering Minor 35**	**Prerequisite Courses—Chemical Engineering Minor** — **Hours** *Choose one of the following:* 5 CHEM 122—General Chemistry I Lecture (4) CHEM 123—General Chemistry Laboratory I (1) **OR** CHEM 116—Honors General Chemistry I (5) CHEM 342—Physical Chemistry I 3 CS 109—C/C++ Programming for Engineers with MatLab 3 MATH 180—Calculus I 4 MATH 181—Calculus II 4 MATH 210—Calculus III 3 MATH 220—Introduction to Differential Equations 3 PHYS 141—General Physics I (Mechanics) 4 PHYS 142—General Physics II (Electricity and Magnetism) 4 **Total Hours—Prerequisites for Chemical Engineering Minor 33**
Required Courses—Chemical Engineering Minor — **Hours** CHE 210—Material and Energy Balances 4 CHE 301—Chemical Engineering Thermodynamics 3 CHE 321—Chemical Reaction Engineering 3 ***Choose one of the following courses:*** 3–4 CHE 311—Transport Phenomena I (3) ME 211—Fluid Mechanics I (4) ***Choose one of the following courses:*** 3–4 CHE 312—Transport Phenomena II (3) ME 321—Heat Transfer (4) CHE 313—Transport Phenomena III (3) **Total Hours—Required Courses for Chemical Engineering Minor 16–18**	**Required Courses—Chemical Engineering Minor** — **Hours** CHE 210—Material and Energy Balances 4 CHE 301—Chemical Engineering Thermodynamics 3 CHE 321—Chemical Reaction Engineering 3 ***Choose one of the following courses:*** 3–4 CHE 311—Transport Phenomena I (3) ME 211—Fluid Mechanics I (4) ***Choose one of the following courses:*** 3–4 CHE 312—Transport Phenomena II (3) ME 321—Heat Transfer (4) CHE 313—Transport Phenomena III (3) **Total Hours—Required Courses for Chemical Engineering Minor 16–18**